The Journal of Open Source Software

ysoisochrone: A Python package to estimate masses and ages for YSOs

Dingshan Deng [1], **Ilaria Pascucci** [1], **and Rachel B. Fernandes** [2,3,4]

1 Lunar and Planetary Laboratory, the University of Arizona, Tucson, AZ 85721, USA **2** President's Postdoctoral Fellow **3** Department of Astronomy & Astrophysics, 525 Davey Laboratory, The Pennsylvania State University, University Park, PA 16802, USA **4** Center for Exoplanets and Habitable Worlds, 525 Davey Laboratory, The Pennsylvania State University, University Park, PA 16802, USA

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Warrick Ball

Reviewers:

- @heloises
- @vsquicciarini

Submitted: 04 October 2024
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Background and Methods

There has been a long history of estimating stellar age and masses from the stellar evolutionary models (e.g., Siess et al. (2000), Feiden (2016), Baraffe et al. (2015)). Different methods have been employed, from finding the closest track to an object's luminosity and temperature (e.g., Manara et al. (2023)) to employing a Bayesian approach which enables estimating uncertainties on the inferred ages and masses (e.g., Jørgensen & Lindegren (2005), Gennaro et al. (2012), Andrews et al. (2013)). Our primary method is a Bayesian inference approach, and the `Python` code builds on the `IDL` version developed in Pascucci et al. (2016). The code estimates the stellar masses, ages, and associated uncertainties by comparing their stellar effective temperature ($T_{\rm eff}$), bolometric luminosity ($L_{\rm bol}$), and their uncertainties with different stellar evolutionary models, including those specifically developed for YSOs. As assumed in previous literature, we adopt uniforms priors, and the likelihood functions are defined following:

$$\mathcal{L}(T_i, L_i | T_{\rm obs}, L_{\rm obs}) = \frac{1}{2\pi\sigma_{T_{\rm obs}}\sigma_{L_{\rm obs}}} \exp\left(-\frac{1}{2}\left[\frac{(T_{\rm obs}-T_i)^2}{\sigma^2_{T_{\rm obs}}} + \frac{(L_{\rm obs}-L_i)^2}{\sigma^2_{L_{\rm obs}}}\right]\right), \quad (1)$$

where the T and L are the effective temperature and bolometric luminosity, respectively. T_i, L_i are the values from the evolutionary model grids, and $T_{\rm obs}, L_{\rm obs}$ are the observed values for each target with their uncertainties ($1\,\sigma$ from the assumed Gaussian distribution) described as $\sigma_{T_{\rm obs}}, \sigma_{L_{\rm obs}}$.

Our method also uses a combination of the pre-main-sequence evolutionary tracks from Feiden (2016) and Baraffe et al. (2015) for hot ($T_{\rm eff} > 3,900$) and cool stars ($T_{\rm eff} \leq 3,900$), respectively. This aligns with the choice as initially suggested in Pascucci et al. (2016) to derive the stellar masses of chameleon I young stellar objects (YSOs). `ysoisochrone` also has a new algorithm to find the zero age main sequence (ZAMS) automatically so that post-main sequence tracks are not included when interpolating to a finer grid of evolutionary tracks (e.g., Fernandes et al. (2023)). This algorithm also enables `ysoisochrone` to handle other stellar evolutionary models that are not only focused on pre-main-sequence stars, such as PARSEC tracks (Bressan et al., 2012). User-developed evolutionary tracks can be also utilized when provided in the specific format described in the code documentation.

We also provide two other ways to estimate the stellar masses and ages from these isochrones. - The classical method that finds the closest point from the isochrones for each YSOs based on their $T_{\rm eff}$ and $L_{\rm bol}$ (the uncertainties are ignored in this method). - In some cases, when a good measurement of the stellar luminosity is unavailable, we provide an option to set up the assumed age and then derive the stellar mass. Some examples when this method is useful include: targets that are very young and exceptionally bright; and targets with an edge-on disk so that the stellar $L_{\rm bol}$ is significantly underestimated.

Statement of Need

`ysoisochrone` is a `Python3` package that utilizes stellar evolutionary tracks to estimate stellar masses and ages of pre-main sequence stars with a Bayesian framework. While several papers in the literature utilize this method (e.g.,), an open-source code for YSOs is not available. `ysoisochrone` fills this gap. The code provides a uniform platform to handle different evolutionary models along with tutorials and detailed documentation for first users.

Acknowledgements

We thank Greg Herczeg for suggesting the blending of the Feiden and Baraffe evolutionary tracks to handle a large range of stellar masses. D.D. and I.P. acknowledge support from Collaborative NSF Astronomy & Astrophysics Research grant (ID: 2205870).

References

Andrews, S. M., Rosenfeld, K. A., Kraus, A. L., & Wilner, D. J. (2013). The Mass Dependence between Protoplanetary Disks and their Stellar Hosts. *771*(2), 129. https://doi.org/10.1088/0004-637X/771/2/129

Baraffe, I., Homeier, D., Allard, F., & Chabrier, G. (2015). New evolutionary models for pre-main sequence and main sequence low-mass stars down to the hydrogen-burning limit. *577*, A42. https://doi.org/10.1051/0004-6361/201425481

Bressan, A., Marigo, P., Girardi, Léo., Salasnich, B., Dal Cero, C., Rubele, S., & Nanni, A. (2012). PARSEC: stellar tracks and isochrones with the PAdova and TRieste Stellar Evolution Code. *427*(1), 127–145. https://doi.org/10.1111/j.1365-2966.2012.21948.x

Feiden, G. A. (2016). Magnetic inhibition of convection and the fundamental properties of low-mass stars. III. A consistent 10 Myr age for the Upper Scorpius OB association. *593*, A99. https://doi.org/10.1051/0004-6361/201527613

Fernandes, R. B., Hardegree-Ullman, K. K., Pascucci, I., Bergsten, G. J., Mulders, G. D., Cunha, K., Mamajek, E. E., Pearson, K. A., Feiden, G. A., & Curtis, J. L. (2023). Using Photometrically Derived Properties of Young Stars to Refine TESS's Transiting Young Planet Survey Completeness. *166*(4), 175. https://doi.org/10.3847/1538-3881/acf4f0

Gennaro, M., Prada Moroni, P. G., & Tognelli, E. (2012). Testing pre-main-sequence models: the power of a Bayesian approach. *420*(2), 986–1018. https://doi.org/10.1111/j.1365-2966.2011.19945.x

Jørgensen, B. R., & Lindegren, L. (2005). Determination of stellar ages from isochrones: Bayesian estimation versus isochrone fitting. *436*(1), 127–143. https://doi.org/10.1051/0004-6361:20042185

Manara, C. F., Ansdell, M., Rosotti, G. P., Hughes, A. M., Armitage, P. J., Lodato, G., & Williams, J. P. (2023). Demographics of Young Stars and their Protoplanetary Disks: Lessons Learned on Disk Evolution and its Connection to Planet Formation. In S. Inutsuka, Y. Aikawa, T. Muto, K. Tomida, & M. Tamura (Eds.), *Protostars and planets VII* (Vol. 534, p. 539). https://doi.org/10.48550/arXiv.2203.09930

Pascucci, I., Testi, L., Herczeg, G. J., Long, F., Manara, C. F., Hendler, N., Mulders, G. D., Krijt, S., Ciesla, F., Henning, Th., Mohanty, S., Drabek-Maunder, E., Apai, D., Szűcs, L., Sacco, G., & Olofsson, J. (2016). A Steeper than Linear Disk Mass-Stellar Mass Scaling Relation. *831*(2), 125. https://doi.org/10.3847/0004-637X/831/2/125

Siess, L., Dufour, E., & Forestini, M. (2000). An internet server for pre-main sequence tracks of low- and intermediate-mass stars. *358*, 593–599. https://doi.org/10.48550/arXiv.astro-ph/

DRAFT

0003477

DRAFT